be any taller than the thickness of a sheet of paper for you to exert more energy climbing it than a rock climber on Earth would exert ascending a three-thousand-mile-high cliff. In short, where gravity is high, the high places tend to fall, filling in the low places—a phenomenon that sounds almost biblical, in preparing the way for the Lord: "Every valley shall be raised up, every mountain and hill made low; the rough ground shall become level, the rugged places a plain" (Isaiah 40:4). That's a recipe for a sphere if there ever was one. For all these reasons, we expect pulsars to be the most perfectly shaped spheres in the universe.

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For rich clusters of galaxies, the overall shape can offer deep astrophysical insight. Some are raggedy. Others are stretched thin in filaments. Yet others form vast sheets. None of these have settled into a stable—spherical—gravitational shape. Some are so extended that the fourteen-billion-year age of the universe is insufficient time for their constituent galaxies to make one crossing of the cluster. We conclude that the cluster was born that way because the mutual gravitational encounters between and among galaxies have had insufficient time to influence the cluster's shape.

But other systems, such as the beautiful Coma cluster of galaxies, which we met in our chapter on dark matter, tell us immediately that gravity has shaped the cluster into a sphere. As a consequence, you are as likely to find a galaxy moving in one direction as in any other. Whenever this is true, the cluster cannot be rotating all that fast; otherwise, we would see some flattening, as we do in our own Milky Way.

The Coma cluster, once again like the Milky Way, is also gravitationally mature. In astrophysical vernacular, such systems are said to be "relaxed," which means many things, including the fortuitous fact that the average velocity of galaxies in the cluster serves as an excellent indicator of the total mass, whether or not the total mass of the system is supplied by the objects used to get the average velocity. It's for these reasons that gravitationally relaxed systems make excellent probes of non-luminous "dark" matter. Allow me to make an even stronger statement: were it not for relaxed systems, the ubiquity of dark matter may have remained undiscovered to this day.

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